

MAST90067 <i>Advanced Methods: Transforms*</i>	Semester 1, 2022
MAST90031 <i>Enumerative Combinatorics*</i>	Semester 1, 2021
MAST10007 <i>Linear Algebra</i>	Semester 1, 2018
MAST30028 <i>Numerical and Symbolic Mathematics</i>	Semester 2, 2014
MAST10005 <i>Calculus 1</i>	Semester 1, 2014

The University of Saskatchewan

MATH 327 <i>Graph Theory*</i>	Term 2, 2015-2016
STAT 241 <i>Probability Theory*</i>	Term 1, 2015-2016
MATH 328 <i>Combinatorics and Enumeration*</i>	Term 2, 2014-2015
STAT 241 <i>Probability Theory*</i>	Term 2, 2014-2015

STUDENT & POSTDOC SUPERVISION

- Dr. Chris Bradly (Postdoctoral Researcher, 2024-2025)
- Nathan Britt (MSc, completed in 2025)
- Tim Banova (PhD, co-supervised with Prof. Mark Holmes, completed in 2024)
- James Gleeson (MSc, completed in 2024)
- Xin Huang (MPhil, co-supervised with Prof. Mark Holmes, completed in 2023)
- William Briggs (MSc, co-supervising with Prof. Aleks Owczarek, completed in 2022)
- Ruijie Xu (PhD, co-supervised with Prof. Aleks Owczarek and Dr. Richard Brak, completed in 2021)
- Gordon Li (2024), Zhong Zheng (2023) and William Mead (Summer 2017/18) (SCIE30001 undergraduate research projects)

RESEARCH & EDUCATION INTERESTS

My research interests are in statistical mechanics, particularly lattice models of walks, polygons, animals and trees, and their applications to modelling interacting polymer systems. Problems in these fields are also frequently connected with complex analysis, stochastic processes and algorithms for counting and simulating discrete structures.

I am also interested in the way that changing technology is affecting higher education, and how universities are adapting to the advent of generative AI and mathematical software. The way that students learn is rapidly changing, and the way that we teach and assess mathematics must also evolve.

GRANTS & AWARDS

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| Chief Investigator on Discovery Project (DP) from the Australian Research Council (ARC)
Title: <i>Transformative simulation techniques for complex polymer networks</i>
Budget: \$412,000 | 2023-2026 |
| Discovery Early Career Researcher Award (DECRA) from the ARC
Title: <i>Statistical mechanics and topology of polymer systems</i>
Budget: \$337,869 | 2017-2020 |
| Gavin Brown Prize from the Australian Mathematical Society
for best paper published in 2008-2016 (see [P7] in the attached Publication List) | 2018 |

PUBLICATIONS

30+ articles published in journals in mathematics, physics, and computer science; 6 conference papers; 1 book chapter. More than 30 collaborators. For a full list see my website www.nicholasbeaton.com.

CONFERENCE ORGANISATION

- Co-organiser of *GuttmannFest 2025: A conference in celebration of Tony Guttmann's 80th birthday*
June 30 – July 2, 2025
University of Melbourne, Australia
- Co-organiser of Special Session on *Topology, geometry and combinatorics of biopolymers*
Joint AustMS / NZMS / AMS Conference 2024
December 9-13, 2024, Auckland NZ
- Co-organiser (Proceedings Editor) of the 31st International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2019)
July 1-5, 2019
University of Ljubljana, Ljubljana, Slovenia

Co-organiser (Proceedings Editor) of the 29th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2017)
July 9–13, 2017
Queen Mary University of London, London, UK

Co-organiser of three Contributed Minisymposia on *Combinatorics, topology and statistical mechanics of polymer models*
11th Biennial Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM)
June 1–4, 2015
University of Saskatchewan, Saskatoon, Canada

SELECTED INVITED PRESENTATIONS *Polymer Modelling and DNA Topology*
November 9–14, 2025, Banff International Research Station, Canada

Monte Carlo Algorithms in Statistical Mechanics
June 26–July 7, 2023, MATRIX, Creswick, VIC

A Tour of Combinatorics and Statistical Mechanics in Memory of Richard Brak
February 7, 2022, The University of Melbourne, VIC

Invited Seminar
September 14, 2021, Université Paris 13, France (online)

Transcendence et Combinatoire
January 8, 2021, Université de Versailles-St Quentin, France (online)

The Topology of Nucleic Acids: Research at the Interface of Low-Dimensional Topology, Polymer Physics and Molecular Biology
March 25–29, 2019, Banff International Research Station, Canada

Invited Seminar
June 27, 2018, Colorado State University, Fort Collins, USA

Conference on Means, Methods and Results in the Statistical Mechanics of Polymeric Systems II
June 12–14, 2017, Fields Institute, Toronto, Canada

Monash Workshop on Self-interacting Processes
August 29–31, 2017, Monash University, VIC

Knots and Polymers: Aspects of topological entanglement in DNA, proteins and graph-shaped polymers
August 8–10, 2017, Ochanomizu University, Tokyo, Japan

OUTREACH, ENGAGEMENT & LEADERSHIP Member of the *Research and Industry Committee* in the School of Mathematics and Statistics at the University of Melbourne.

Secretary of the Australian and New Zealand Association for Mathematical Physics (ANZAMP, a division of AustMS), 2024-2026.

Member of the Advisory Board for *Journal of Physics A: Mathematical and Theoretical*.

Reviewer for ARC Discovery Project proposals.

Mentor for several students at Elizabeth Blackburn Sciences (University High School) doing Extended Investigation research projects in 2019 and 2020.

Volunteer for *MathsCraft* events, which connect mathematicians with high school students and teachers at events focused on open-ended mathematical problems and thinking like a researcher.

Host of the School Maths Olympics at the 2022 University of Melbourne Open Day.